ENVIRONMENTAL AUDIT, NEERI KOLKATA ZONAL CENTRE

The National Environmental Engineering Research Institute (NEERI), Kolkata Zonal Centre

(KZC) commissioned a comprehensive Environmental Audit covering water use, solid waste

management, and energy consumption — as mandated by the Central Pollution Control

Board (CPCB) framework for institutional environmental management.

METHODOLOGY:

A structured three-phase approach was adopted,

- **Pre-Audit:** General and detailed questionnaires distributed to gather baseline data on energy, water, and waste.
- **Site Visit :** Physical verification of infrastructure - old/new buildings, water distribution network, electrical panels, HVAC/AC units, lighting, garden areas, and waste collection.
- **Data Analysis:** Utility bill review, equipment inventory, occupancy schedules, and environmental performance benchmarking.

AUDIT SCOPE & APPROACH

Water Audit	Waste Audit	Energy Audit
Metered & unmetered consumption, loss quantification, efficiency measures	Source inventory, waste characterisation, collection & treatment planning	Utility analysis, equipment assessment, conservation measures

KEY FINDINGS:

Energy - High per-capita electricity consumption: ~10 units/person/day — significantly

above benchmark.

Equipment - AC units and ancillary equipment showing reduced efficiency due to deferred

maintenance.

Lighting - Widespread use of old tube lights and legacy fittings with poor energy

performance.

RECOMMENDATIONS:

- **Replace** all tube lights and legacy fittings with energy efficient LED systems.
- **Implement** decentralized switching for outdoor lighting to eliminate waste.

- **Schedule** regular preventive maintenance for all AC units and electrical equipment.
- **Adopt** sub-metering to track and reduce per-capita energy consumption.
- **Establish** a centralized waste segregation and collection mechanism at KZC.
- **Optimize** water distribution network to identify and plug unmetered losses.
- **Install** rainwater harvesting and water recycling systems where feasible.

BENEFITS DELIVERED:

- Regulatory compliance mapping against CPCB norms.
- Identification of energy & cost savings hotspots.
- Third-party verified environmental performance data.
- Capacity building for KZC technical staff.
- Actionable database for future plant upgrades.

Impact Highlight

Potential to reduce electricity consumption by20–30% through LED retrofits, decentralized switching, and preventive maintenance — delivering significant cost savings and reducing the carbon footprint of KZC operations.